

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) have achieved remarkable progress across generative tasks such as image and video synthesis, yet they continue to produce hallucinated samples (*hallucinations*) that conflict with real-world knowledge—for instance, an implausible duplicate cup floating beside another cup. Such failures are pervasive, and the absence of a feasible methodology for systematically quantifying them has both impeded progress on the problem and obscured principled pathways toward next-generation generative models that are governed by factual constraints. We close this gap by isolating a well-defined and especially tractable form of hallucination, which we term **counting hallucination**: the generation of an incorrect number of instances or structured objects—such as a hand rendered with six fingers—despite such patterns being entirely absent from the training data. To enable its rigorous study, we construct **CountHalluSet**, a dataset suite equipped with explicit, unambiguous counting criteria and spanning a controlled spectrum of morphological complexity through three datasets: ToyShape, SimObject, and Real-Hand. Building on these datasets, we establish a standardized evaluation protocol for quantifying counting hallucinations and use it to systematically characterize how the principal sampling conditions of DPMs—solver type, ODE solver order, number of sampling steps, and initial noise—govern the resulting hallucination levels. A correlation analysis against widely used perceptual metrics, including the Fréchet Inception Distance (FID), further establishes that such metrics fail to capture counting hallucinations consistently, demonstrating that perceptual fidelity and counting-based factuality constitute distinct axes of generative quality.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become the dominant paradigm in generative modeling, producing high-quality samples across a broad range of modalities, including text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Despite these advances, diffusion models remain unable to faithfully capture the factual properties of their training data, and they routinely generate outputs that violate consistency with real-world knowledge. A characteristic and readily observable manifestation of this failure is the generation of objects with an incorrect number of spatially intricate components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

A growing body of work has sought to identify and characterize the common failure modes of text-guided diffusion models. Borji (2023) systematically catalogued a range of qualitative shortcomings in text-to-image models, while Liu et al. (2024) introduced an adversarial search procedure to surface undesirable behaviors and failure cases in text-to-image generation, and Wu et al. (2023) developed an iterative method to discover and mitigate spurious correlations. These efforts, however, remain largely qualitative, and none delivers a quantitative account of the factual violations that persist even in visually convincing outputs.

More recently, *hallucination* has emerged as a distinct and consequential failure mode of diffusion models, attracting sustained attention from the research community (Aithal et al., 2024). It differs fundamentally from the failure modes discussed above. Text-prompt-based failures can be assessed by measuring the alignment between input conditions and outputs (e.g., CLIP score (Hessel et al.,

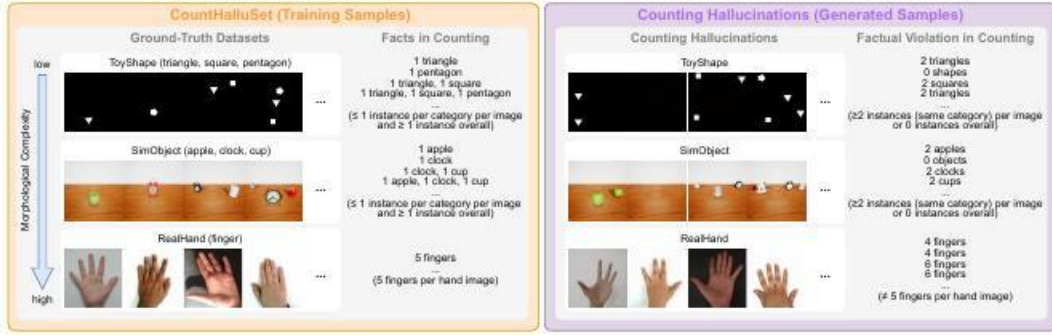


Figure 1: Illustrative comparison between the proposed **CountHalluSet** training data (left) and the **counting hallucinations** produced by models trained on it (right). Each dataset defines objects governed by explicit counting criteria: in ToyShape and SimObject, every image contains at most one instance per category and at least one instance overall, whereas in RealHand every image depicts exactly five fingers. Counting hallucinations are generated images that violate these dataset-specific counting rules—for example, two apples within a single image, or an image with a plausible background but no objects at all—configurations that never occur in the corresponding training data (here, SimObject).

2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)), and conspicuous degradations such as severe distortions, blur, and artifacts are trivially detectable; hallucinations, by contrast, typically appear visually plausible and arise independently of the conditioning signal, manifesting as subtle violations of factual consistency with respect to the training data—incorrect object counts, geometries that look plausible yet are physically impossible, or other breaches of physical law. Precisely because these violations are fine-grained, they pose an especially demanding challenge for current AI systems, which frequently fail to detect them: even state-of-the-art large vision-language models such as GPT-4o struggle with a task as elementary as counting the fingers in an unambiguous hand image. Establishing a standardized, well-defined protocol for reliably quantifying hallucinated samples is therefore essential—not merely as a measurement tool, but as the foundation for a principled path toward next-generation generative models that satisfy strict factual consistency, a capability that is indispensable to their role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the common categories of hallucinated content—counts, colors, spatial relationships, geometry, and physical realism—counting errors are uniquely well suited to systematic study: they are simultaneously easy to identify and directly quantifiable, and they therefore furnish an objective, unambiguous target for measurement. This tractability motivates our decision to ground the investigation in a precisely delimited form of hallucination, which we term **counting hallucination**. In this setting, the admissible number of objects in the training data is explicitly predetermined; consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values. An image depicting a hand with six fingers, for example, is classified as a counting hallucination because such a configuration never occurs in the RealHand dataset, as shown in the last row of Fig. 1. To enable the controlled study of this phenomenon, we construct the dataset suite CountHalluSet, comprising three datasets that deliberately span a spectrum of morphological complexity for the countable objects: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 presents representative examples from these datasets alongside their corresponding counting hallucinations.

The linchpin of our quantification protocol is the accurate identification and enumeration of objects, which we accomplish through counting models tailored to each dataset. Equipped with these datasets and counting models, we systematically examine how the principal denoising conditions of diffusion models—the number of sampling steps, the initial noise, the order of the ODE solver, and the solver type—govern counting hallucination rates, and we further quantify the correlation between counting hallucination rate and perceptual metrics, including the Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our study yields the following principal contributions:

- We show that the numerical strategies commonly relied upon in diffusion models—most notably increasing the number of sampling steps from 25 to 100—fail to mitigate counting hallucinations when generating human hands, even though the same strategies are demonstrably effective on simpler synthetic datasets, revealing that added sampling computation does not translate into counting-based factuality.
- We establish that the counting hallucination rate is only weakly correlated with perceptual metrics, whereas the non-counting failure rate is strongly correlated with them—demonstrating that widely used perceptual metrics such as FID track perceptual degradation but are largely blind to counting-based factual violations.
- Building directly on these findings, we introduce joint-diffusion models, a novel training paradigm that substantially reduces both counting hallucinations and non-counting failures on the RealHand dataset.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Motivated by the intense scrutiny of hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), the study of hallucinations in unconditional diffusion models has attracted increasing interest. Notably, Aithal et al. (2024) advanced a qualitative hypothesis attributing hallucinated samples to interpolation between distinct data modes. Valuable as this insight into potential causes is, the field still lacks a systematic, quantitative analysis of hallucinations in diffusion models—precisely the gap our work is designed to close.

We therefore develop a methodology for the accurate identification and quantification of hallucinations in diffusion-generated outputs. Such quantification does more than measure a symptom: it deepens the understanding of these models’ undesirable behaviors and makes explicit how the magnitude of denoising error—as modulated by the number of sampling steps or the order of the ODE solver—governs the severity of hallucinations.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are the two most widely used metrics for evaluating text-guided image generation. Yet evaluation methods built on language-based models are fundamentally ill-suited to assessing hallucinated images. Two independent lines of evidence establish this limitation. First, there exists a statistical lower bound on the propensity of pretrained language models to hallucinate certain fact types, which in turn caps the detection accuracy attainable by language models alone (Kalai & Vempala, 2024). Second, a persistent performance gap separates human annotators from LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023). Together, these results indicate that language-model-based scoring cannot serve as a reliable arbiter of factual consistency.

A complementary family of quantitative metrics—FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021)—is routinely employed in image generation, all of them operating on feature embeddings extracted from a pretrained Inception-v3 model (Szegedy et al., 2016). *Whether these metrics can capture the severity of hallucinations in generated datasets*, however, remains an open question—one that this work resolves from the perspective of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $x_0 \sim q_0(x_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (equivalently, the *diffusion process*) that progressively perturbs the data by injecting Gaussian noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample x_t conditioned on the original data

$\mathbf{x}_0$ is given by:

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_t\}_{t=1}^T$ is a predefined, t -dependent noise schedule, most commonly instantiated as a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The factor $\bar{\alpha}_t$ governs the signal-to-noise ratio at step t , interpolating between $\bar{\alpha}_0 = 1$ (no noise) and $\bar{\alpha}_T \approx 0$ (nearly pure noise). An equivalent reparameterization of Eq. (1) is:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the marginal distribution $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ converges to a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$; in widely used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021), T is conventionally set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (equivalently, the *denoising process*) of DPMs iteratively reconstructs the observed data from the diffused sample $\mathbf{x}_T$. Concretely, a neural network $\epsilon_\theta(\mathbf{x}_t, t)$ —the *noise prediction model*—is trained to predict the noise ϵ from the noisy sample $\mathbf{x}_t$, with its parameters θ optimized by minimizing the objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, samples from the data distribution $q_0(\mathbf{x}_0)$ are obtained by reversing the diffusion process over the same number of steps, a procedure known as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1 - \beta_t}} (\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where the sole unknown term $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ through the noise prediction model, so that an equivalent formulation of Eq. (3) is:

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Whereas the ancestral sampling of Eq. (4) incorporates a stochastic term (Kloeden et al., 1992), it admits an associated deterministic counterpart expressible as an ordinary differential equation (ODE), the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau) \mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2} \beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2} \beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and originates from ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Discretizing the time axis as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$, and applying the variation-of-constants formula for the exact solver over a single step yields:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$, and $\Delta t > 0$ denotes the step size, which permits ODE-based solvers to accelerate sampling by taking larger steps than ancestral sampling, so that the total number of sampling steps equals $T/\Delta t$. Because the linear part $\mathcal{G}(t_2, t_1)$ admits a closed-form solution, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical effort—through higher solver orders and finer step counts—on accurately approximating the nonlinear integral term involving h . A more complete theoretical analysis is provided in Appendix A.1 through A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section develops the methodology and datasets that make counting hallucinations rigorously measurable. We first formalize the notion of counting criteria and the counting hallucinations defined as deviations from them, then introduce the three datasets of the CountHalluSet suite. For each dataset we specify its counting criteria and the corresponding counting hallucinations, and detail the standardized protocol by which they are quantified.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We now give a general definition of the counting criteria that a training dataset imposes, and characterize counting hallucinations as violations of these predefined constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the count of category- c objects, $N_c(\mathbf{x})$, is constrained to lie in a predefined admissible set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$); that is, $N_c(\mathbf{x}) \in \mathcal{S}_c$. We additionally require that every valid reference sample contain at least one object, i.e., $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of categories under consideration.

Counting hallucinations as deviations from the counting criteria. A generated sample $\hat{\mathbf{x}}$ is said to exhibit a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ precisely when both of the following conditions hold jointly:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must attain sufficient visual quality for the background and any present objects to be clearly discernible. This precondition isolates counting hallucinations from other prominent failure modes, most notably severe degradations, so that the two are never conflated. We encode it through a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, the *counting-ready indicator*, where 1 signals that the image quality suffices for counting and 0 signals otherwise. In practice, this indicator may be instantiated via human annotation, via a classifier calibrated against human annotations, or via other automated criteria.
2. **Violation of counting facts.** A violation of counting facts occurs whenever some category c satisfies $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or the image contains no objects at all ($\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$), the latter being disallowed because the dataset requires at least one object per sample. In practice, we obtain the evidence for such violations from a counting model that predicts object counts in generated images.

A counting hallucination (CH) is thus declared exactly when the image is of sufficient quality for counting ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts deviate from the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite comprises three datasets purpose-built for the study of counting hallucinations—ToyShape, SimObject, and RealHand—which together span a controlled progression from simple synthetic shapes to real-world human hands. Representative examples from each dataset, together with their corresponding counting hallucinations, appear in Fig. A1 and A2, and additional statistical summaries are reported in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset consists of 30,000 images built from three geometric shapes—*triangle*, *square*, and *pentagon*—each rendered in white with an identical area of 120 pixels and positioned so that no two shapes overlap within an image. Its counting criteria stipulate that each image contain at most one instance of each shape category and at least one shape in total, as

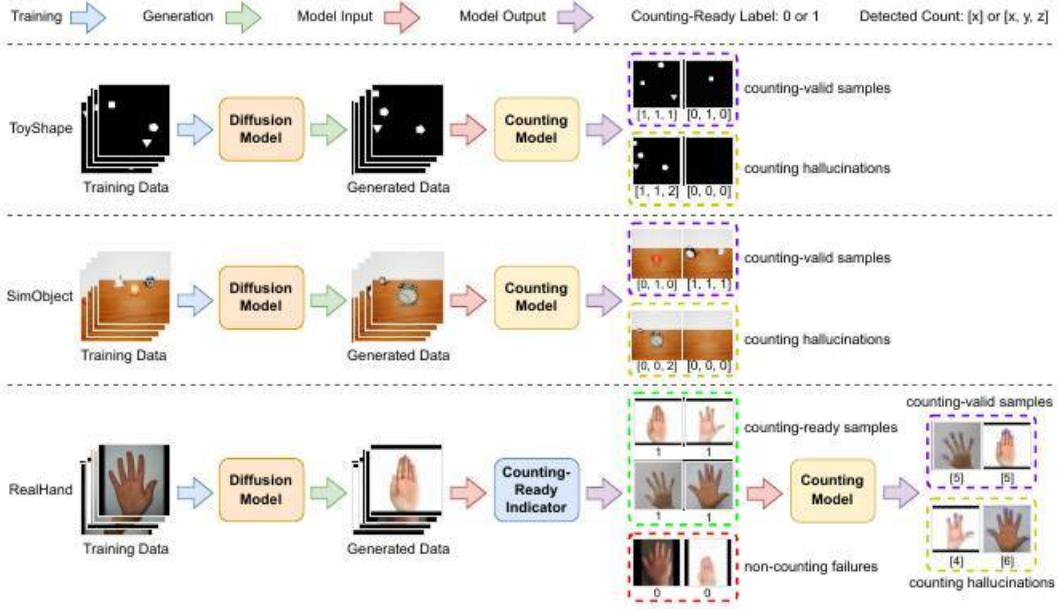


Figure 2: The evaluation procedure of the CountHalluSet suite. For ToyShape and SimObject, generated images are assessed for counting directly. For RealHand, a counting-ready indicator—a binary classifier aligned with human judgments—is first applied to separate counting hallucinations from non-counting failures such as severely deformed fingers, after which counting proceeds. Count labels correspond to ToyShape: *triangle, square, pentagon*; SimObject: *mug, apple, clock*; RealHand: *finger*.

illustrated in the top-left of Fig. 1. Formally, $C = \{\text{triangle, square, pentagon}\}$, and for each class $c \in C$ we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under the counting criteria defined above, any image generated by a model trained on ToyShape that contains two or more shapes of the same category constitutes a counting hallucination—for instance, the image with two pentagons in the top-right region of Fig. 1. Empty images—those displaying only the black background with no shapes—likewise qualify as counting hallucinations. Further examples are provided in Appendix A.5.

Counting model used in ToyShape. We identify counting errors with a dedicated counting model and dispense with the counting-ready indicator, since the dataset’s simplicity renders severely degraded images negligibly rare. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on over 400k ToyShape images in which each category appears 0–3 times per sample (i.e., $S_c = \{0, 1, 2, 3\}$), a design that deliberately spans the full range of plausible configurations—both counting-valid samples and counting hallucinations—so as to support accurate quantification. The resulting model attains over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. Given a diffusion model trained on ToyShape, we generate a sample set matched in size to the training data (30,000 images), apply the counting model to predict the number of objects in each generated image, and flag every image that violates the counting criteria as a counting hallucination, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset comprises 30,000 rendered images of everyday objects drawn from three categories—*mug, apple, and clock*—each realized in 10 distinct intra-class variations. Objects are placed at random on a wooden tabletop under fixed lighting, with minimal mutual occlusion, as shown in the middle-left of Fig. 1. The dataset is synthesized in Unreal Engine 5 (Epic Games, 2024), whose photorealistic rendering faithfully simulates object geometry, material attributes, lighting, and reflections. Its counting criteria parallel those of ToyShape (see Sec. 4.2.1): formally, $C = \{\text{mug, apple, clock}\}$, and for each class $c \in C$ we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Owing to the close correspondence of their counting criteria, the definition of counting hallucinations, the construction of the counting model, and the quantification protocol all follow the ToyShape procedure exactly, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. To study counting hallucinations in a realistic setting, we construct the RealHand dataset of 5,050 human hand images aggregated from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow², from which we discard images exhibiting blur or artifacts. Each retained image depicts a five-fingered human hand, spanning both dorsal (back) and palmar (front) views under varied lighting conditions and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria require exactly five fingers per image: $C = \{finger\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, so that $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

Counting hallucinations and non-counting failures. The generation of realistic hand images is intrinsically susceptible to failures beyond counting hallucinations (e.g., extra or missing fingers)—including blurriness, severely deformed fingers, and visual artifacts—which must be separated out before counting can be meaningful. We accomplish this separation with a counting-ready indicator, implemented as a binary classifier. Examples of both counting hallucinations and non-counting failure samples are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To disentangle counting hallucinations from the aforementioned failure modes, we fine-tune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images as a counting-ready indicator aligned with human judgment, and separately fine-tune a YOLO-12 (Tian et al., 2025) detector on 2k fingertip-annotated generated images to perform the counting itself. Evaluated against three-annotator human labels on 500 randomly sampled images, the indicator and the counting model achieve over 96% and 99% accuracy, respectively, establishing both as reliable instruments for the RealHand protocol. Representative YOLO detection results are shown in Appendix A.6.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, where the simplicity of the data and the rarity of severe failure cases (e.g., distorted or unrecognizable objects) permit all generated images to be evaluated directly, RealHand requires an additional filtering stage. Concretely, we first apply the counting-ready indicator to retain only hand images of sufficient visual quality, and only then invoke the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape we train a DDPM (Ho et al., 2020) from scratch, whereas for SimObject and RealHand we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on CelebA-HQ (Karras et al., 2018). Across all settings the number of timesteps is fixed at $T = 1,000$ under a standard linear noise schedule, and images are resized to 128×128 for the DDPM and to 256×256 for the LDMs. Optimization uses Adam (Kingma, 2015) with a learning rate of 0.0001 and an effective batch size of 256, run for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively.

Inference of Diffusion Models. Ancestral sampling (i.e., DDPM) is run for the full 1,000 steps. For the ODE-based solvers we consider both a first-order and a second-order method: DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)) as the first-order solver, and DPM-Solver-2 (Lu et al., 2022) as the second-order solver. Each solver is evaluated under the three most widely used denoising-step configurations—25, 50, and 100 steps—so that solver order and step count are varied independently. On RealHand, the counting model employs a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless stated otherwise, the number of generated samples matches the number of training samples in every evaluation, and all reported results are averaged over three random seeds to ensure statistical stability.

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

Table 1: Counting hallucination rates (CHR; the number of counting hallucinations divided by the total number of generated samples) under different solver configurations and initial-noise settings across the three datasets ToyShape, SimObject, and RealHand. For RealHand we additionally report the non-counting failure rate (NCFR)—the fraction of failure cases captured by the counting-ready indicator—and the total failure rate (TFR), defined as the sum of CHR and NCFR. *Diffused* and *Normal* denote the ground-truth initial noise and standard Gaussian noise, respectively. Lower values indicate better performance throughout.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across the full grid of denoising conditions—solver type, ODE solver order, number of sampling steps, and initial noise setting—for all three datasets (ToyShape, SimObject, and RealHand). Five consistent patterns emerge.

(1) **Increasing the number of sampling steps reduces counting hallucinations on synthetic datasets but aggravates them on RealHand.** As Table 1 shows, across the standard ODE solver configurations (25, 50, and 100 steps), additional sampling steps systematically lower the counting hallucination rate on ToyShape and SimObject, yet raise it on RealHand—the sole exception being 100 steps with DPM-Solver-2. This dissociation indicates that the simpler, more structured distributions of the synthetic datasets genuinely benefit from finer solver granularity, whereas the substantially more complex distribution of RealHand causes additional steps to overfit local inconsistencies and thereby amplify hallucinations.

(2) **Higher-order ODE solvers markedly reduce non-counting failures yet aggravate counting hallucinations on RealHand at standard step counts.** Comparing DPM-Solver-1 with DPM-Solver-2 on RealHand at the standard sampling steps (e.g., 25 and 50) in Table 1 exposes a consistent trade-off: the higher-order solver substantially lowers the non-counting failure rate while simultaneously driving the counting hallucination rate upward, and the same trade-off recurs as the number of sampling steps is increased for either solver. The overarching conclusion is unambiguous: allocating more computation—i.e., adopting “better” sampling conditions—reliably improves perceptual quality (lower non-counting failure rates) but does nothing to enforce the factual correctness of finger counts.

(3) **DDPM decisively outperforms the ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across every comparison, ancestral sampling (i.e., DDPM) attains the lowest CHR, NCFR, and TFR, thereby furnishing a practical lower bound on the failure rates of these generative models. This establishes ancestral sampling as the most effective of the evaluated strategies for minimizing hallucinations.

(4) **A more informed initial-noise configuration reduces both counting hallucinations and non-counting failures.** As Table 1 shows, “diffused” noise (ground-truth initial noise) consistently yields lower CHR, NCFR, and TFR than “normal” noise (standard Gaussian noise). This gain arises because diffused noise more closely matches the initial-noise distribution encountered during training, demonstrating that a better-aligned initialization mitigates not only counting hallucinations but non-counting failures as well.

(5) **Greater morphological complexity of the countable objects systematically increases counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand reveals a monotonic increase in counting hallucination rate with the morphological complexity of the target objects. The fine-grained detail and intricate spatial relationships intrinsic to morphologically complex objects

Table 2: Correlation of the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with the perceptual metrics FID and MUSIQ. The PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset; FID for PTI-Hand is computed between the RealHand images and the model-generated samples. Here, r denotes the p-value assessing the statistical significance of the correlation r or ρ , and $\dagger$ marks the setting in which DDPM results are integrated with those from the various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01	0.87	0.01	-0.97	<0.01	0.20	0.69
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—	0.49	0.14	-0.53	0.13	—	—
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29	0.84	<0.01	-0.97	<0.01	-0.14	0.78
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—	0.75	0.01	-0.51	0.15	—	—
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—	0.89	—	0.97	—	0.47	—
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—	0.82	—	0.74	—	—	—

thus impose a fundamental burden on the ability of diffusion models to preserve accurate object counts, precipitating more frequent additions or omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Because a counting hallucination is a form of factualness hallucination, it is natural to ask: *does a better perceptual metric—say, a lower FID—necessarily entail fewer counting hallucinations, on the grounds that it reflects a smaller distributional gap between generated and training data?* Our evidence answers this question decisively in the negative: it does **not**.

To probe this relationship, we conduct a correlation analysis of CHR against perceptual metrics and, in parallel, of NCFR against the same metrics. We consider two complementary and widely used perceptual measures—the distribution-based FID (Heusel et al., 2017) and the instance-based MUSIQ (Ke et al., 2021)—and quantify dependence with three estimators that together span linear, monotonic, and general nonlinear relationships: the Pearson correlation (Lee Rodgers & Nicewander, 1988) (linear dependence), the Spearman rank correlation (Spearman, 1904) (monotonic dependence), and the Distance Correlation (Székely et al., 2007) (general nonlinear dependence). Two findings, detailed below, jointly substantiate our conclusion.

✦ **The correlation between CHR and perceptual metrics is dataset- and solver-dependent rather than intrinsic.** As the first and third rows of Table 2 show, the Pearson correlation between CHR and FID changes sign across datasets—positive for SimObject, negative for RealHand, and positive again for PTI-Hand—while the corresponding Spearman correlations move from positive to negative to negative. The Distance Correlation likewise declines from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), indicating that increasingly complex generative content progressively weakens any association between CHR and FID; integrating the DDPM results with those from the ODE solvers attenuates these correlations still further. No stable, transferable relationship between CHR and perceptual quality therefore exists.

✦ **The correlation between NCFR and perceptual metrics is uniformly strong and positive.** In sharp contrast to the weak and unstable CHR correlations, the correlation between NCFR and perceptual metrics is strong and robust across every setting. For NCFR versus FID, in particular, all correlation coefficients exceed 0.88 with very small p-values, confirming high statistical significance. Taken together, the two findings demonstrate that perceptual metrics faithfully track non-counting (perceptual) failures while remaining largely blind to counting-based factual violations.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

Having established our findings in a controlled setting, we now demonstrate that they extend to a large-scale pretrained text-to-image model. Concretely, we evaluate Stable-Diffusion-3.5-Medium (Esser et al., 2024) with the very same counting-ready indicator and counting model introduced in Sec. 4.2.3, so that the protocol is transferred without modification (see Appendix A.8 for details). As Figure 3 shows, the manner in which solver type and sampling steps modulate counting hallucinations

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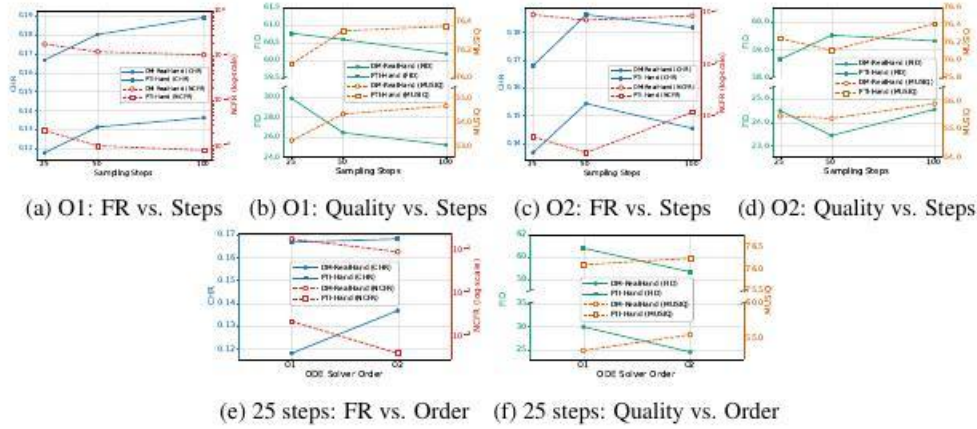


Figure 3: Trend-consistency comparison between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote the first- and second-order ODE solvers (DPM-Solver-1 / DPM-Solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). The failure-rate (FR) metrics comprise CHR and NCFR; the quality metrics comprise FID and MUSIQ. Across variations in both sampling steps and solver order, the two models exhibit highly consistent trends, corroborating that our controlled RealHand findings extend to large-scale pretrained text-to-image models.

in this large pretrained model aligns closely with the trends revealed by our controlled RealHand experiments. This close agreement confirms the validity of the findings in Sec. 5.2 and establishes the broad applicability of our evaluation protocol. The correlation results in Table 2 reinforce the same conclusion: on the pretrained model, we again observe weak correlations between CHR and perceptual metrics and strong correlations between NCFR and perceptual metrics. Collectively, these results attest to the robustness and generality of our findings and show that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (Observation 5, Sec. 5.2) indicate that diffusion models capture counting-based factuality more reliably when the underlying visual structure is morphologically simple (e.g., masked objects). This observation motivates a concrete design question: if a diffusion model were to denoise general visual representations jointly with an explicit, easily modeled structural constraint—such as an object mask that encodes basic topology and counts—within a shared latent space, could this joint learning paradigm suppress counting hallucinations? We term models that jointly learn general visual representations and structural constraints during denoising **joint-diffusion models** (JDM).

Concretely, we concatenate the latent representation of the hand mask channel-wise with that of the corresponding RGB image, as shown in Fig. 4, and then fine-tune the CelebA-HQ-pretrained UNet (Karras et al., 2018)—following the RealHand configuration of Sec. 5.1—on the RealHand images together with their associated masks, which are extracted using SAM-2 (Ravi et al., 2024). To accommodate both modalities, we expand the input and output channels of the denoising UNet to match the combined latent dimensionality of the image and mask latents (i.e., $3 + 3 = 6$ channels), while leaving all intermediate layers unchanged.

As Table 3 shows, JDM consistently achieves lower failure rates than the vanilla LDM—on both CHR and NCFR—across all sampling conditions, demonstrating its effectiveness at accurately generating the structurally intricate objects that are most prone to counting ambiguities. JDM does, however, incur a higher FID. This trade-off is attributable to the increased denoising burden of jointly modeling two modalities under a small-scale training regime, which leads the model to exchange a measure of perceptual quality for improved structural realism—precisely the axis of factuality that perceptual metrics fail to reward (Sec. 5.3). We further observe promising reductions in counting hallucinations when JDM is applied to large-scale pretrained diffusion models under carefully controlled fine-tuning settings. Additional details, results, and discussion are provided in Appendix A.9.

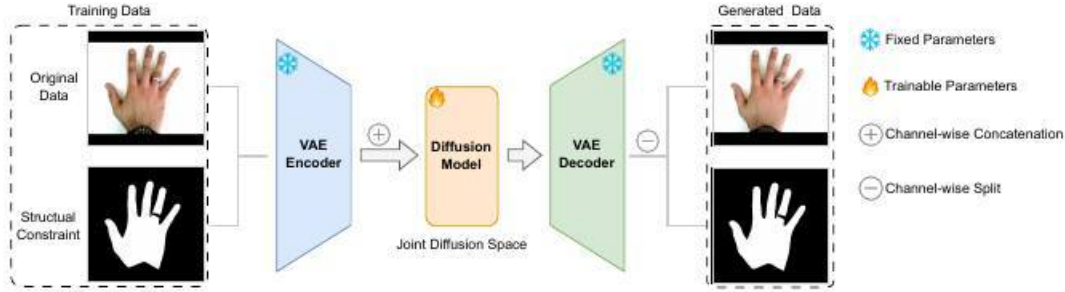


Figure 4: Diagram of the proposed joint-diffusion model (JDM). The model concatenates, channel-wise, the latent representations of the hand RGB image and its corresponding segmentation mask. By denoising both modalities jointly, this learning paradigm encourages anatomically plausible hand poses and correct finger topology during generation, yielding reduced CHR and NCFR.

Table 3: Failure-rate comparison between the vanilla LDM and our proposed JDM on the RealHand dataset across the different ODE solvers and sampling steps. Lower values indicate better performance. Although JDM jointly generates hand RGB images and segmentation masks, its FID is computed solely on the RGB outputs to preserve comparability with the baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

6 CONCLUSION

This work takes the first systematic step toward quantifying hallucinations in diffusion models and, in doing so, opens new avenues for improving factual generation. Centering the study on counting hallucinations, we introduce **CountHalluSet** together with a practical, reproducible methodology for measuring them. Through comprehensive and controlled experiments, we establish that the numerical strategies commonly relied upon fail to mitigate counting hallucinations when generating human hands, and our correlation analyses demonstrate that widely used perceptual metrics such as FID and MUSIQ cannot consistently reflect the severity of counting hallucinations—showing that perceptual fidelity and counting-based factuality are distinct objectives. Experiments on large-scale pretrained text-to-image models corroborate both the generality and the validity of these findings, and the joint-diffusion paradigm we introduce charts a concrete and promising path toward enforcing factual consistency in image generation.

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A APPENDIX

A.1 PROOF OF THE BOUND ON THE TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the model reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence between them is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Exploiting the Markov property, both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$ decompose as follows:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule to decompose the KL divergence then yields:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalizing—that is, discarding a subset of the random variables—cannot increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality holds strictly only in the ideal case $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$ —precisely the case in which the diffusion model recovers the underlying data distribution $q(\mathbf{x}_0)$ exactly. Excluding the data entropy $H(\mathbf{x}_0)$, this upper bound on the overall KL divergence $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$ between the true and model-generated data coincides with the variational bound derived in DDPM (Ho et al., 2020), which establishes that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between the true and model-generated distributions.

Diffused-prior gap. The first source of error is the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the prior distribution $p_\theta(\mathbf{x}_T)$ used at sampling time. This mismatch is a direct consequence of the number of diffusion steps T being finite in practice. The discrepancy is formally quantified through a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence, though other suitable metrics such as FID or the Wasserstein distance could equally be used to evaluate this distributional discrepancy.

Present from the very outset of sampling, the diffused-prior gap induces a deviation that propagates throughout the denoising process and ultimately manifests as a persistent output error.

Noise-model prediction error. The second source of error stems from the noise prediction model: its estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , since the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022) guarantees approximation but not exact recovery. Writing $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ for this prediction error, Eq. (5) shows that any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse step reuses the mean computed at the previous iteration, this bias accumulates along the denoising trajectory and thereby contributes to the overall prediction error.

A.2 PROOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

We build on Eq. (7) and take a one-step numerical solver as the running example. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and originates from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Correspondingly, let $\hat{\mathbf{x}}_k$ denote the numerical trajectory computed by an ODE solver at the discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ denote the estimated model function. Our goal is to characterize the total accumulated trajectory error of a specific numerical trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$. Defining the step- k trajectory error as $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$, we analyze the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Applying the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}) gives:

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$, and the integration limits run backward in time.

Numerical path evolution. The one-step numerical solver approximates this solution, and its numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution at τ_{k+1} of the ODE driven by $\hat{h}$ and initialized at $\hat{\mathbf{x}}_k$ at time τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method additionally introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12) yields the recurrence:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) [\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u)] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. Unrolling this recurrence from $k = 0$ to $N - 1$ —with $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$ and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$ —and applying it repeatedly gives:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ then produces:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial-error term captures the deterministic contribution of a specific initial-error instance to the final path error, the statistical origin of which lies in the distributional diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As given in Eq. (2), the forward process of DDPM (Ho et al., 2020) is formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

and the corresponding reverse process is defined as:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), whereas its deterministic reverse process (with $\eta = 0$) is defined as:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share an identical forward (diffusion) process, DDIM is not obtained merely by stripping the stochastic term from DDPM’s reverse (denoising) process; the two methods employ genuinely different coefficient formulations for the predicted-noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. As Table A1 shows, each dataset is balanced across its constituent categories, and Table A2 further confirms that the distribution of images over distinct object counts is comparable, so that no object-count bias is introduced across datasets. The datasets appear in the same order in both tables, and the category ordering in Table A1 is as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Number of objects per category for each dataset, confirming a balanced category distribution.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images grouped by object count for each dataset, confirming a comparable distribution over object counts.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide representative examples of counting hallucinations for all three datasets in Fig. A2. For ToyShape and SimObject, counting hallucinations either contain more than one instance of at least one category or contain no instance at all while retaining a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting-hallucinated samples are characterized by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images of the first row in Fig. A2(c) appear at first glance to display five fingers; on closer inspection, however, one finger is partially incomplete and visibly artifacted, so that only four valid fingers remain, and these samples are therefore classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative examples of YOLO predictions on the RealHand dataset. The YOLO detector accurately localizes fingertips across a wide range of hand poses and orientations, spanning both dorsal and palmar views. It nonetheless fails to flag certain anatomically implausible configurations—such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Because these samples still contain exactly five detected fingertips, they are counted as five-finger hands and remain indistinguishable from anatomically valid ones on the basis of finger count alone.

Figure A3(b) shows several counting-hallucinated samples for which YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips). Figure A3(c) further illustrates several rare edge cases (<1%) in which the detector fails on fingertips with unclear boundaries, cracked fingers, or floating artifacts. Such cases can be partially mitigated by tuning YOLO’s inference hyperparameters (e.g., the IoU and confidence thresholds), but doing so risks degrading detection on the overwhelming majority of counting-valid samples. Because a single, consistent detection configuration is applied across every RealHand evaluation, the presence of these rare cases does not affect the validity of our overall conclusions, and their resolution remains a challenging, low-reward endeavor best left to future work.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of the three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the RealHand evaluation procedure first applies the counting-ready indicator to separate counting-ready samples (i.e.,

counting-valid and counting-hallucinated) from images unsuitable for counting, the latter typically exhibiting severe visual artifacts such as distorted, cracked, partially visible, or floating fingers. Counting-ready samples are then further partitioned into counting-valid and counting-hallucinated according to whether the number of detected valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

For the PTI-Hand experiments, we prompt Stable-Diffusion-3.5-Medium with 10 distinct hand-generation prompts, listed below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Our definition of hallucination presupposes that the model generates patterns never observed during training—an assumption that is difficult to verify for large-scale pretrained diffusion models, whose full training distribution is inaccessible. To keep the quantification of counting hallucinations feasible in this setting, we adopt a deliberately conservative criterion, treating any generated hand image with six or more fingers as counting-hallucinated. This criterion is well justified: whereas real images may depict fewer than five visible fingers because of occlusion, articulation, or viewpoint, real hands with six or more fingers are exceedingly unlikely. A finger count of ≥ 6 therefore constitutes a reliable and empirically verifiable signal of counting hallucination. Generated examples are shown in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

For the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, compared with 80k steps for the vanilla LDM. This larger training budget is necessary because extending the latent space from three to six channels introduces a substantial mismatch with the pretrained latent distribution of the LDM, and additional optimization is required to close this distributional gap so that JDM’s modeling capacity can be fully realized.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

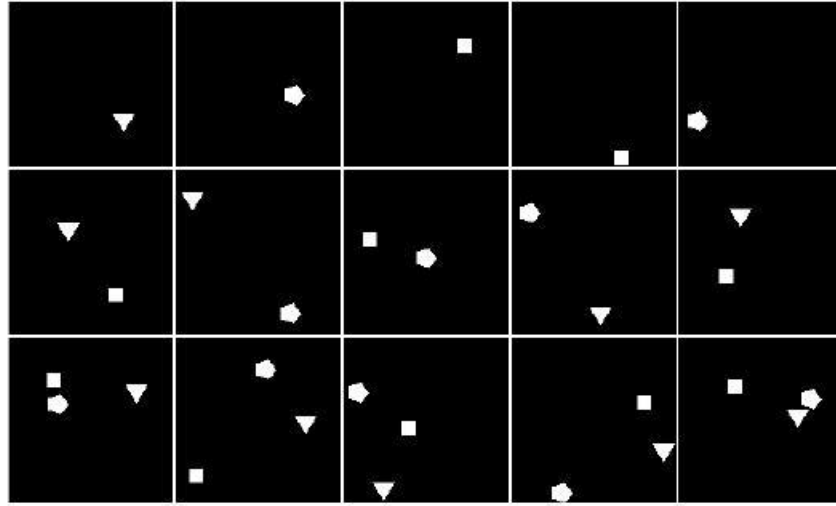
Importantly, JDM is best understood as a **novel training paradigm** for improving factual generation in diffusion models, rather than as a superior inference strategy. On the one hand, because JDM extends only the input and output channels of the UNet while leaving the dimensions of all intermediate layers unchanged, it **introduces negligible inference-time overhead relative to the vanilla LDM**. On the other hand, augmenting a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and perturbs the original denoising distribution**, which renders direct fine-tuning of such a model nontrivial.

Table A3: Comparison of failure rates and image quality on RealHand between fine-tuning SD-1.5 and fine-tuning SD-1.5 under the joint-diffusion training paradigm. The first row (baseline) reports results from fine-tuning Stable-Diffusion-1.5 on RealHand data; the second row reports results obtained by freezing all intermediate layers of the baseline and fine-tuning only the input and output layers with extended channel dimensions. Lower values indicate better performance.

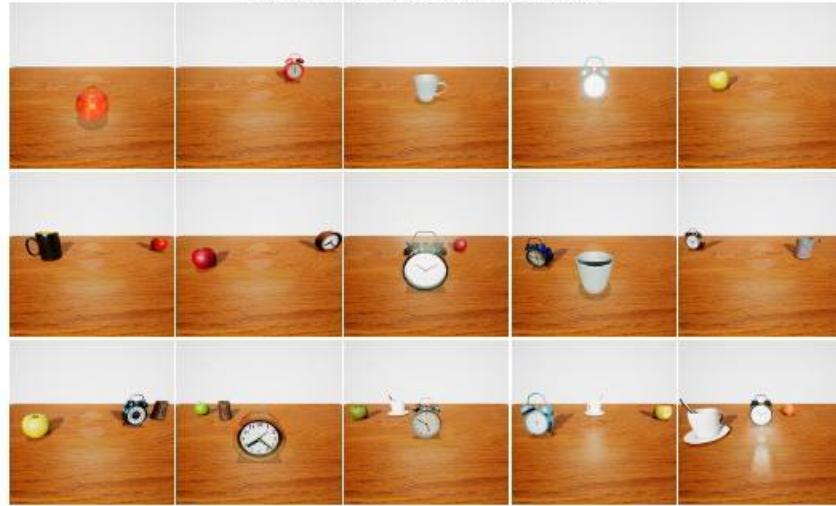
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

Despite this challenge, we still observe promising reductions in counting hallucinations when JDM is applied to large-scale pretrained diffusion models under carefully controlled fine-tuning conditions. In practice, aligning JDM’s extended latent space with the highly rigid latent geometry of large pretrained models proves difficult at the limited scale of RealHand. To strike a practical balance, we adopt a two-stage fine-tuning strategy: we first fine-tune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and fine-tune only the input and output layers with extended channels. As Table A3 shows, even under this heavily constrained setup the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence presented in this work establishes JDM as a promising direction for enforcing factual consistency in image generation, and **characterizing its behavior and performance under genuine large-scale training remains an important and compelling avenue for future research.**



(a) Examples in the ToyShape dataset

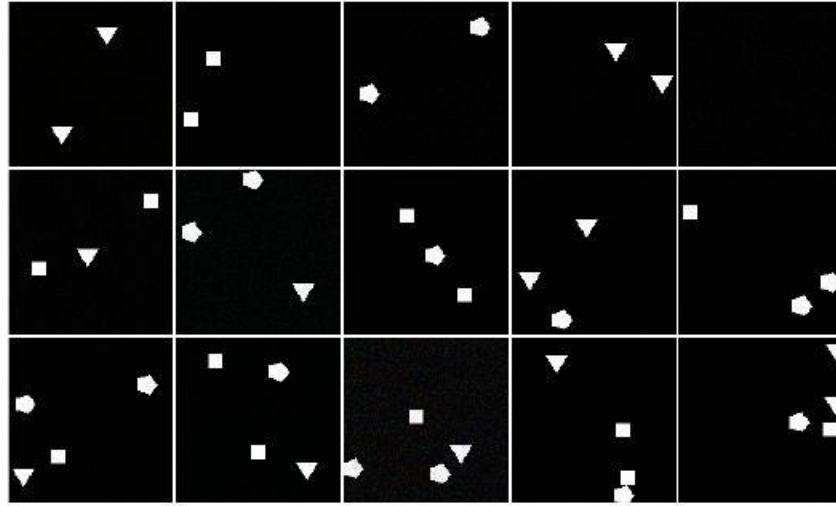


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

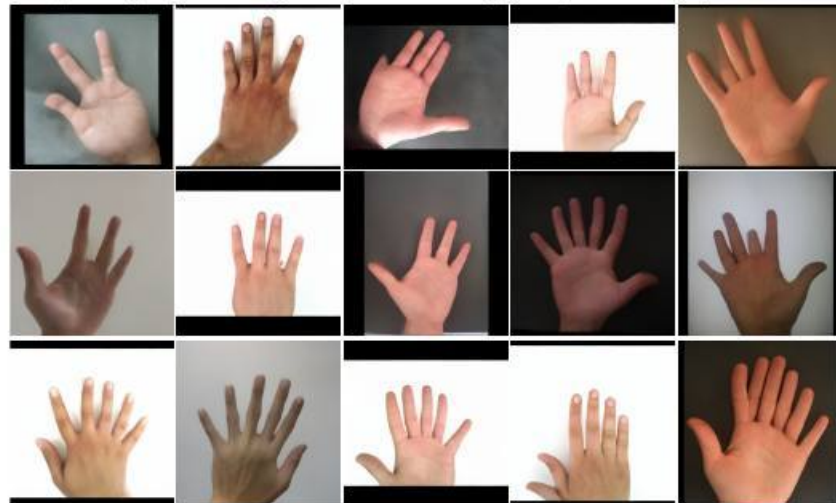
Figure A1: Representative examples from the three datasets of the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

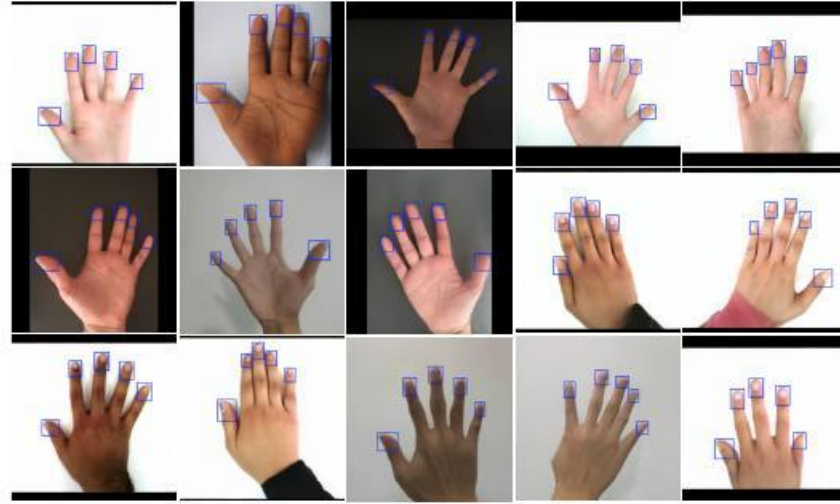


(b) Examples of counting hallucinations corresponding to the SimObject dataset

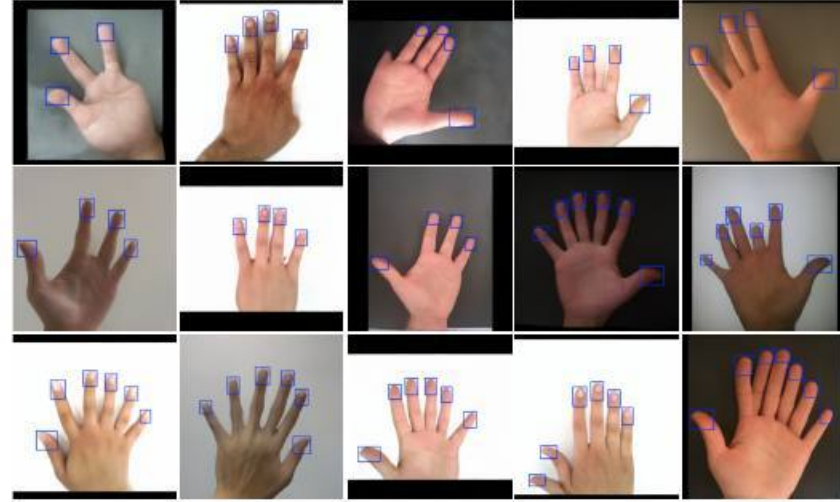


(c) Examples of counting hallucinations corresponding to the RealHand dataset

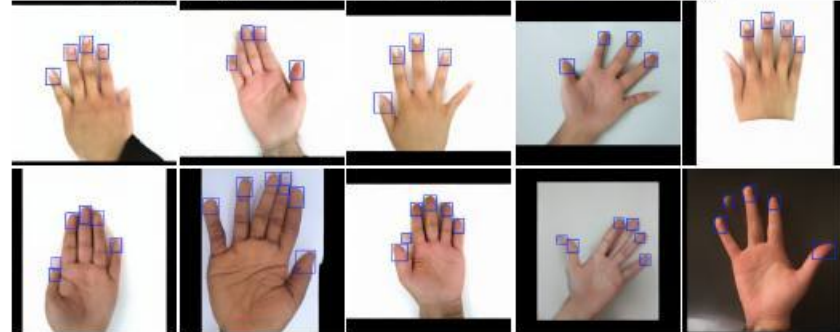
Figure A2: Representative counting hallucinations for the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers



(b) Counting-hallucinated YOLO predictions with not five fingers



(c) Rare edge cases where YOLO predictions fail

Figure A3: Examples of YOLO predictions on generated RealHand images. (a) Counting-valid samples for which YOLO correctly predicts five fingertip bounding boxes (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip-count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Rare edge cases that the counting-ready indicator fails to filter out and that YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.

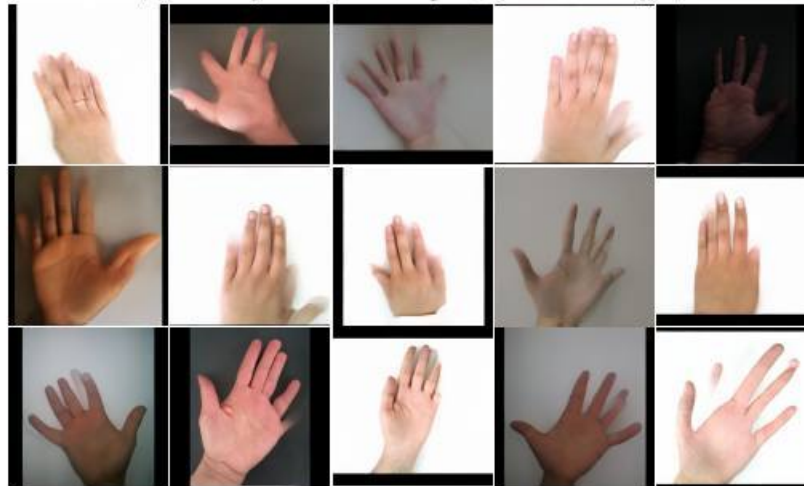
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(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of the three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting owing to severe visual artifacts such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of the three classes of generated hand images produced by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting owing to severe visual artifacts such as distorted, cracked, partially visible, or floating fingers.